

CORPUS-BASED SPEECH EVALUATION

AfriSpeech Benchmark Report

Final evaluation of baseline transcription, controlled noise and local-only inference.

Sauti Yetu Project | 27 clips | Nine metadata groups | Four run exports

Scope of this report

The supplied results contain 27 AfriSpeech corpus clips with English reference text. This report presents the measured baseline, noise and local-only results from the completed benchmark runs.

Principal findings

Baseline transcription. Intron Sahara has the lowest mean WER at **41.77%**, followed by Meta MMS at **57.61%** and OpenAI Whisper at **66.17%**. Sahara leads overall, but the ranking varies across the three-clip metadata groups. [B]

Noise robustness. The three-model sweep contains **324/324 scored results**, with no recorded errors. At 5 dB, mean WER is **54.28%** for Sahara, **85.08%** for MMS and **96.52%** for Whisper. Sahara is lowest at all four tested SNR levels. [N]

Local execution. The separate local-noise sweep has **216/216** scored results; the offline pass has **54/54**. Offline mean WER is **57.61%** for MMS and **68.38%** for Whisper. These results establish completion of the local paths, not independently verified network isolation. [L, O]

The report retains the measured scores, distinguishes each run, and explains the reference and configuration limitations that affect interpretation. It is a descriptive benchmark of the recorded configurations, not a general ranking of model families or a clinical validation.

Source keys: [B] baseline, 13:45:46 UTC; [N] three-model noise, 13:59:49 UTC; [L] local-noise, 13:59:04 UTC; [O] offline, 14:01:16 UTC. All runs were exported on 15 September 2026. Full provenance is on page 10.

1. Dataset and evaluation design

All four JSON exports contain the same 27 unique filenames, matching metadata and reference strings. The recordings total **242.86 seconds** (approximately 4.05 minutes), with clip durations from **3.18 to 20.99 seconds**. Metadata identifies the device/source as afrispeech-200 corpus and the original noise condition as corpus recording. [B, N, L, O]

The collection contains **14 clinical** and **13 general** clips, with three recordings in each of nine language_pair groups. The supplied references are English; no annotated language switches are present. Group labels such as Swahili-English therefore describe how these clips are grouped, not demonstrated bilingual speech in each recording.

Run	Command suffix	Primary evidence	Scored coverage
Baseline [B]	--tag baseline	Unperturbed ASR results.	27 x 3 = 81
Noise [N]	--noise --tag noise	Three models at 30, 20, 10 and 5 dB.	27 x 4 x 3 = 324
Local noise [L]	--noise --noise-local --tag noise_local	Whisper and MMS noise sweep.	27 x 4 x 2 = 216
Offline [O]	--offline --tag offline	Dedicated local-only inference pass.	27 x 2 = 54

Each export also contains an unperturbed three-model section with 81 scored records. The noise and offline counts above refer only to their dedicated experiment sections. The repeated sections use the same 27 clips; they are not additional independent samples.

Metric contract

Metric	Definition and reporting rule
WER / CER	The manifest specifies jiwer error rates after lowercasing, punctuation removal, whitespace collapse and trimming. Means are unweighted averages of stored clip scores; they are not pooled corpus error rates.
Latency / RTF	Recorded wall-clock seconds and latency divided by audio duration. RTF below 1 means processing faster than the recording duration for that result; it does not guarantee a short interactive delay.
Noise comparison	Higher SNR is cleaner. Noise changes are compared with the unperturbed section of the same export. No missing measurements are interpolated.
Scoring	WER/CER are reported as stored; lower values indicate fewer transcription errors.
Unmeasured fields	No expected topics, urgencies, departments or slots, and zero annotated language boundaries. Null scores are unavailable, not 0% accuracy.

Models: Intron Sahara via remote API; Meta MMS locally with mms-1b-all; OpenAI Whisper locally, with the checkpoint unspecified in the exports. The JSON files provide results and declared normalization, not a complete record of decoder or audio-transform settings.

2. Baseline transcription performance

The dedicated baseline export is the main clean comparison. All three models produced numeric WER, CER and timing values on all 27 clips, with no recorded errors. Here, baseline means the original corpus recording, not a verified noise-free signal. [B]

Model	Mean WER	Mean CER	Mean latency	Mean RTF	RTF < 1	Zero WER
Sahara	41.77%	22.43%	10.25 s	1.4517	14/27	3/27
MMS	57.61%	27.08%	8.02 s	0.9774	20/27	0/27
Whisper	66.17%	41.00%	4.12 s	0.6242	22/27	0/27

Sahara is lowest in both mean WER and mean CER. It has 16 outright per-clip WER wins, Whisper has 3, and MMS has 2; six clips are tied for lowest WER. The export's single best_model label is not used to allocate tied cases.

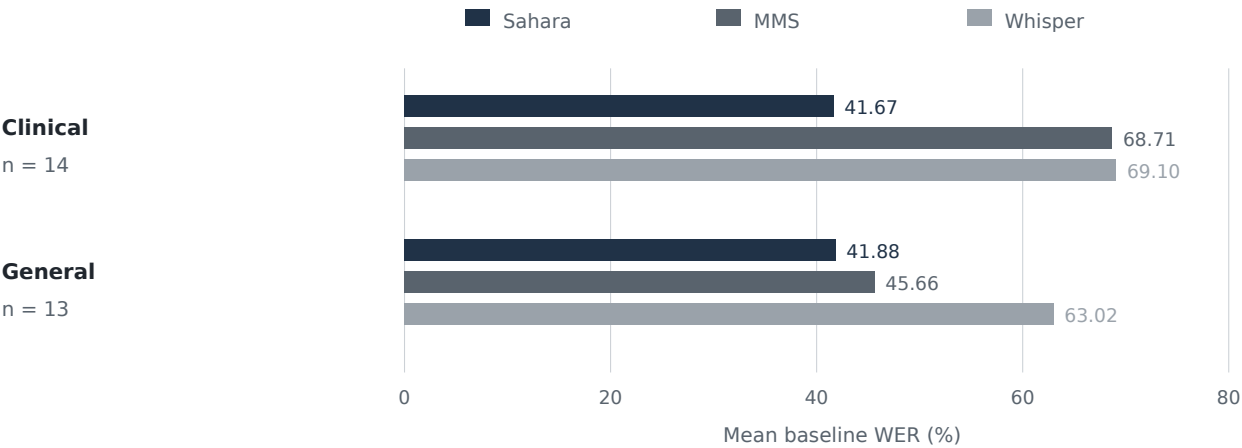


Figure 1. Baseline mean WER by supplied domain. Values are recalculated from the same per-clip scores; clinical n = 14 and general n = 13. Domain labels are descriptive metadata. [B]

What the quality and timing results imply

Sahara has similar mean WER on the clinical and general subsets: **41.67%** and **41.88%**. MMS has a larger gap, with **68.71%** on clinical clips and **45.66%** on general clips. Whisper records **69.10%** and **63.02%**, respectively. The small, unmatched subsets do not isolate clinical vocabulary as the cause of these differences.

Whisper has the lowest mean baseline latency and mean RTF. Its median latency is **1.82 seconds**, compared with **3.53 seconds** for MMS and **8.91 seconds** for Sahara. These timing comparisons combine local and remote paths and are specific to the recorded environment and workload.

Mean RTF can conceal slow individual results: MMS averages 0.9774, but only 20 of 27 clips are below 1. Sahara averages 1.4517, while 14 of 27 individual clips are below 1. Both the mean and per-clip count are retained to show that distinction.

Interpretation

Sahara is the strongest baseline transcription system in this collection, but a 41.77% mean WER remains substantial. The ranking should not be interpreted as error-free transcription or as evidence of reliable clinical decisions.

3. Performance across metadata groups

The table retains the exact language_pair labels from the export. Each row contains only three English-reference clips. It is a descriptive group comparison, not a test of nine demonstrated code-switching patterns. [B]

Export group label	n	Sahara WER	MMS WER	Whisper WER
Swahili-English	3	22.50%	45.05%	88.63%
Hausa-English	3	33.40%	78.39%	56.05%
Yoruba-English	3	59.38%	38.16%	65.08%
Igbo-English	3	17.42%	37.16%	42.37%
Zulu-English	3	61.75%	48.62%	48.92%
Afrikaans-English	3	25.58%	52.85%	112.34%
Luganda-English	3	69.71%	81.22%	56.62%
Kinyarwanda-English-French	3	69.31%	52.15%	46.79%
Pidgin-English	3	16.88%	84.89%	78.74%

The overall ranking is not uniform

Sahara has the lowest group mean WER in **five of nine groups**: Swahili, Hausa, Igbo, Afrikaans and Pidgin. MMS is lowest in Yoruba and Zulu; Whisper is lowest in Luganda and Kinyarwanda. These are descriptive group means, not statistically established population differences.

The Zulu group illustrates a close local-model result: MMS records **48.62%** WER and Whisper **48.92%**. A 0.30 percentage-point difference across three clips is too limited to support a broad preference for one system on that accent.

Language and accent configuration need separate treatment

Metadata language_code values follow the group labels (for example, sw for the Swahili group), while the reference text is English. The export does not show the exact language argument passed into each decoder. Some Whisper outputs depart markedly from the English reference. This makes language-hint handling a configuration question to verify, not a demonstrated cause of the errors.

The accent_country field also contains within-group variation: one Yoruba record is tagged US while two are tagged NG; one Zulu record is tagged LS while two are tagged ZA. These metadata differences reinforce that the groups are not controlled country-level samples. Speaker IDs and demographic strata are not included.

Code-switching boundary

No clip contains a switch-point annotation and every stored switch-point WER is null. This benchmark can describe transcription against these English references; it cannot quantify performance at language switches or establish the code-switching capability of the three models.

4. Controlled-noise transcription

The three-model noise sweep is complete: 27 clips x four SNR levels x three models = **324 scored records**, all without a recorded error. Every chart point uses n = 27. Higher SNR is cleaner; lines connect the four measured conditions only. [N]

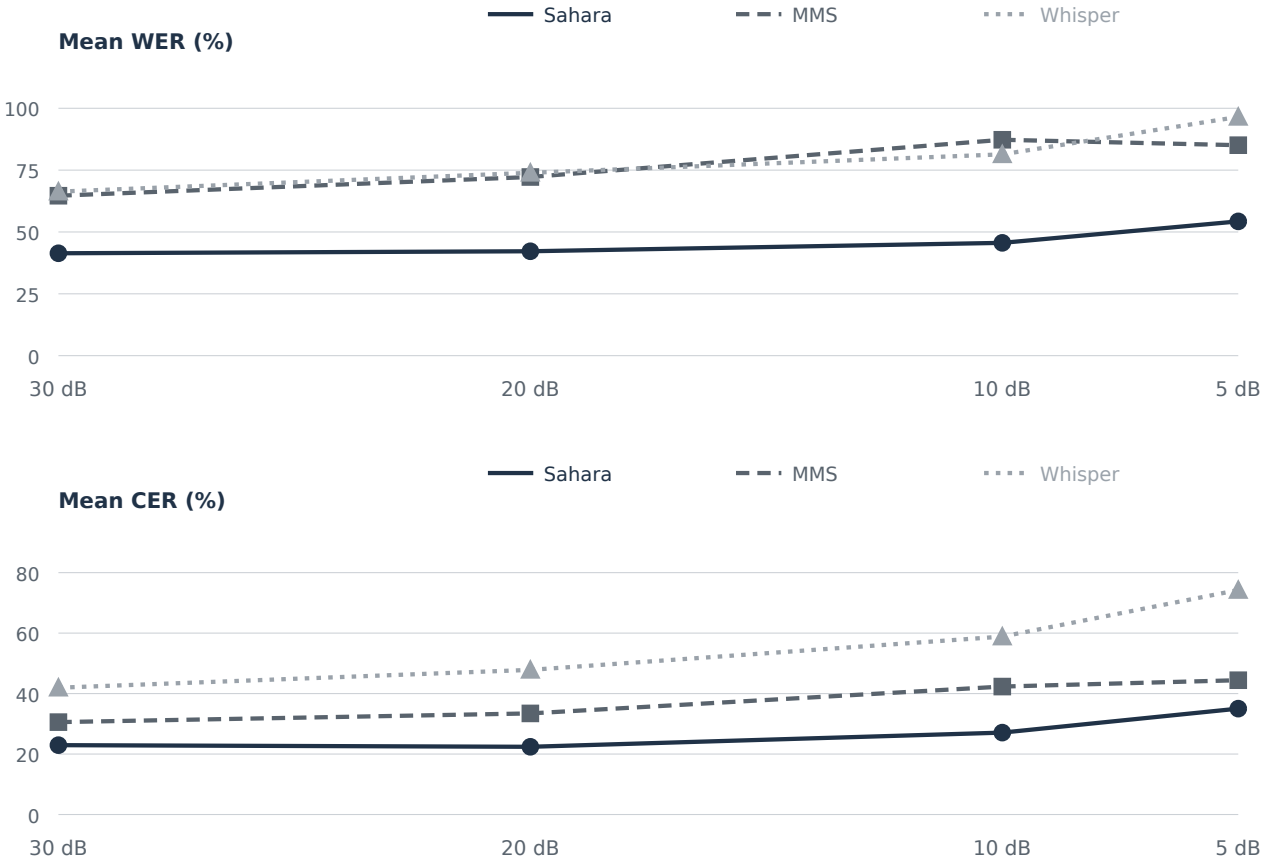


Figure 2. WER and CER across measured SNR levels. Line styles and markers identify the models in grayscale. Both metrics use the same 27 clips at each level.

SNR	Sahara WER	Sahara CER	MMS WER	MMS CER	Whisper WER	Whisper CER
30 dB	41.39%	22.94%	64.65%	30.56%	66.30%	41.94%
20 dB	42.20%	22.40%	72.22%	33.44%	73.92%	47.84%
10 dB	45.60%	27.09%	87.27%	42.32%	81.38%	58.85%
5 dB	54.28%	35.02%	85.08%	44.42%	96.52%	74.25%

Sahara has the lowest WER and CER at every SNR. Its WER increases from **41.39%** at 30 dB to **54.28%** at 5 dB. The same-export unperturbed value is 41.77%; the small 30 dB improvement is descriptive and does not establish a beneficial noise effect.

MMS is lower than Whisper on CER at every SNR. On WER it is lower at 30, 20 and 5 dB, but higher at 10 dB (**87.27%** versus **81.38%**). MMS WER is non-monotonic from 10 to 5 dB, even though its CER increases.

The unperturbed WER values within [N] are Sahara 41.77%, MMS 57.61% and Whisper 68.11%. At 5 dB these become 54.28%, 85.08% and 96.52%. These same-export comparisons avoid treating repeated baseline runs as interchangeable.

5. Noise timing and local-model rerun

Mean RTF in the three-model sweep is shown with the number of clips below RTF 1. The model paths have different local and remote workloads, and the export does not isolate the causes of timing variation. [N]

SNR	Sahara RTF	Sahara < 1	MMS RTF	MMS < 1	Whisper RTF	Whisper < 1
30 dB	0.7701	20/27	0.7355	21/27	1.6178	22/27
20 dB	0.4247	26/27	0.6673	24/27	1.2560	20/27
10 dB	0.4179	26/27	0.6671	23/27	1.1787	19/27
5 dB	0.2969	27/27	0.6603	23/27	1.5647	14/27

Sahara's mean RTF falls as SNR falls in this run; this is a recorded timing pattern, not proof that added noise accelerates inference. Whisper's 30 dB mean of 1.6178 includes a slow result for `afriSpeech_sw_0.wav` (82.12 seconds; RTF 21.2416), although 22 of 27 clips are below RTF 1.

Separate local-noise sweep

The `--noise-local` run contains **216 scored records** for MMS and Whisper only, with no recorded errors. Its top-level baseline still contains all three models; Sahara is absent specifically from the `noise_robustness` section. [L]

SNR	MMS WER	MMS CER	MMS RTF	Whisper WER	Whisper CER	Whisper RTF
30 dB	64.65%	30.56%	0.8878	68.13%	43.02%	0.8774
20 dB	72.22%	33.44%	0.8766	72.35%	45.69%	0.8985
10 dB	87.27%	42.32%	0.8725	81.67%	59.37%	0.6631
5 dB	85.08%	44.42%	0.8701	96.53%	97.07%	1.0864

MMS transcripts and WER/CER values match the three-model sweep in all **108** corresponding cases, although timing differs. Whisper matches in 60 of 108 cases. The two exports are kept separate; these repeated or differing outputs do not independently establish the noise-generation or caching behavior.

Why the two Whisper CER curves differ

At 5 dB, Whisper has similar mean WER in [N] and [L] (96.52% and 96.53%) but different mean CER (74.25% and 97.07%). In [L], `afriSpeech_sw_0.wav` contains a long repeated-character output and records 770.69% CER. That scored outlier contributes to the local-run mean and is retained.

An empty transcript is not automatically an execution error: each Whisper noise run contains one empty output at 10 dB and one at 5 dB, with numeric error scores and `error = null`. These cases remain in the 27-clip denominators.

6. Offline execution and run reconciliation

The `offline_simulation` section contains 27 results each for MMS and Whisper, all completed without a recorded error. Sahara has no offline result. This is the dedicated local-only comparison from `results.offline.json`. [O]

Model	Mean WER	Mean CER	Mean latency	Mean RTF	RTF < 1
MMS	57.61%	27.08%	9.52 s	1.0753	13/27
Whisper	68.38%	39.81%	5.84 s	0.9346	22/27

MMS has **14 outright WER wins**, Whisper has **10**, and **3 clips tie**. MMS has lower mean WER and CER, while Whisper has lower mean latency and RTF. Neither local model has a zero-WER clip in this pass.

Local model	[O] unperturbed WER	[O] offline-pass WER	What is observed
MMS	57.61%	57.61%	Mean WER is unchanged.
Whisper	63.45%	68.38%	Mean WER differs between the two passes.

The supplied chart labels these passes Online (clean) and Offline (clean). Both evaluated models are described as local in the JSON. The WER difference therefore cannot be assigned to network availability without execution details. The report uses unperturbed pass and offline pass to describe the recorded evidence.

Repeated unperturbed sections are not pooled

Export	Sahara WER	MMS WER	Whisper WER
[B] baseline	41.77%	57.61%	66.17%
[N] noise	41.77%	57.61%	68.11%
[L] noise_local	41.77%	57.61%	63.00%
[O] offline	41.77%	57.61%	63.45%

Sahara and MMS retain the same unperturbed mean WER across the four exports. Whisper ranges from 63.00% to 68.11%. The designated baseline [B] is used for the main ranking, while within-run comparisons use each experiment's own unperturbed values. These four passes are not a controlled repeated-trial estimate.

The JSON does not document OS-level network blocking, model-download/cache state, hardware, or cold-start loading. Completion of the local-only pass supports an execution claim; it does not establish independently verified air-gapped operation or full offline operation of the end-to-end application.

7. Results synthesis

What the completed measurements show

Across the measured transcription tasks, Sahara has the lowest baseline WER and remains the lowest-WER system at every tested SNR level. The noise curves show increasing error as the signal becomes noisier, with the clearest separation at 5 dB.

For local execution, MMS has the lower offline mean WER, while Whisper is faster on the recorded offline pass. The timing results are workload-specific and should be treated as engineering observations rather than universal speed claims.

Evidence boundary

This report focuses on the ASR, noise and local-execution measurements directly populated in the result exports. The results are descriptive and should not be interpreted as clinical validation.

8. Data quality and final interpretation

Reference conventions affect the measured error

The references contain forms such as *Theapplication*, *productneeded*, *cappedvenous* and *Managementl n*. In `afrispeech_sw_1.wav`, the reference uses *First-dose* and Sahara outputs *1st-dose*, producing 10% WER. The declared punctuation/whitespace normalization does not resolve every numeral, word-joining or dictated-punctuation convention. [B]

These references are retained exactly as supplied. All 27 filenames and reference strings are unique. The audio was not independently audited, so no reference correction or claim of independently validated annotation quality is made. Unlike the primary PDF's historical bootstrap description, these exports do not identify Sahara as the source of their references.

Limits of the completed experiments

Limitation	Effect on interpretation
Small corpus	Twenty-seven short recordings, with three clips per group, are descriptive evidence. No demographic balance, speaker-count verification, confidence intervals or significance tests are established.
Language versus accent	English references and empty switch-point annotations cannot establish code-switching performance. Decoder language hints need explicit documentation.
Noise recipe	SNR values are present, but the noise source/generator, seed, gain normalization and exact mixing method are absent. The curves do not identify a real clinic noise environment.
Timing context	The exact Whisper checkpoint, hardware, device utilization, warm/cold state and timing boundaries are not fully recorded. Run-to-run speed differences cannot be assigned to a single cause.

The noise export includes silence diagnostics: mean silence ratio **37.72%**, median **37.25%**, and range **15.16%-80.19%**. Mean pause count is **27**; mean longest pause is **0.61 seconds**. The manifest defines silence as 20 ms frames below -40 dBFS. These are recorded diagnostics, not independent validation of SNR mixing. [N]

Final conclusion

Sahara has the lowest overall baseline error and leads the complete three-model noise sweep at every tested SNR. MMS is the stronger local model by overall and offline mean WER, although Whisper leads some groups and the 10 dB WER comparison. Both local paths complete their dedicated runs. The results finalize this corpus benchmark while leaving reference quality and configuration reproducibility as explicit limits.

Appendix A. Sources and verification

Key	Raw source file	Export time (UTC)	Role
[B]	results.baseline.json	13:45:46	Designated baseline ASR results.
[N]	results.noise.json	13:59:49	Complete three-model noise sweep.
[L]	results.noise_local.json	13:59:04	Separate local-model noise sweep.
[O]	results.offline.json	14:01:16	Dedicated local-only pass and its paired unperturbed section.

All exports use schema 3.0, Python 3.13.5 and Windows-11-10.0.26200-SP0, and report commit 854d040f73e464ff903f6d05a059b0474f0c9eaf. They were exported on 15 September 2026. The metrics describe those recorded configurations; the full runtime environment is not captured.

Checks performed for this report

All reproduced overall and group WER/CER, latency and RTF aggregates were recalculated from the stored per-result values and agree within the four-decimal rounding tolerance. All expected models and SNR levels are present. The four unperturbed sections contain 324 scored records in total; the dedicated noise, local-noise and offline sections add 324, 216 and 54, respectively. None records an execution error.

Numeric scores on empty transcripts remain valid scored outcomes. Means are not recomputed after excluding difficult clips, and no missing benchmark result is interpolated. This review checks the supplied metrics and their aggregation, not new transcription from audio or independent edit-distance rescoring.

Derived result	Calculation
Clip wins	Lowest WER per clip; an outright win requires a unique minimum. Tied minima are reported separately.
Group / domain means	Arithmetic mean of the stored per-clip WER values within the supplied metadata group. Domain means are newly derived.
RTF < 1 counts	Number of scored records with stored rtf below 1, divided by the full 27-clip denominator.

Appendix B. Complete baseline score audit

All 27 baseline filenames and stored WER/CER values are included below. Values are percentages; lower is better. The same files are used in the other three exports. Source: [B], clips[].benchmark.results[].

Filename	Sahara WER	Sahara CER	MMS WER	MMS CER	Whisper WER	Whisper CER
afrispeech_sw_0.wav	22.22	20.69	22.22	12.07	100.00	79.31
afrispeech_sw_1.wav	10.00	3.66	60.00	19.51	60.00	18.29
afrispeech_sw_2.wav	35.29	18.35	52.94	19.27	105.88	93.58
afrispeech_ha_0.wav	6.67	1.05	33.33	21.05	20.00	16.84
afrispeech_ha_1.wav	18.52	8.56	51.85	10.36	48.15	20.72
afrispeech_ha_2.wav	75.00	30.77	150.00	103.85	100.00	96.15
afrispeech_yo_0.wav	35.71	12.05	35.71	20.48	78.57	53.01
afrispeech_yo_1.wav	9.09	3.05	45.45	22.14	50.00	25.19
afrispeech_yo_2.wav	133.33	27.59	33.33	13.79	66.67	17.24
afrispeech_ig_0.wav	0.00	0.00	12.00	9.30	8.00	7.75
afrispeech_ig_1.wav	13.79	5.53	37.93	19.10	34.48	16.58
afrispeech_ig_2.wav	38.46	26.03	61.54	47.95	84.62	61.64
afrispeech_zu_0.wav	36.36	12.68	63.64	23.94	54.55	23.94
afrispeech_zu_1.wav	88.89	57.14	22.22	15.87	22.22	15.87
afrispeech_zu_2.wav	60.00	32.47	60.00	18.18	70.00	23.38
afrispeech_af_0.wav	0.00	0.00	33.33	18.33	116.67	110.00
afrispeech_af_1.wav	46.43	28.19	46.43	22.87	114.29	85.64
afrispeech_af_2.wav	30.30	9.35	78.79	29.44	106.06	55.14
afrispeech_lg_0.wav	53.57	50.00	21.43	9.26	14.29	7.41
afrispeech_lg_1.wav	66.67	18.57	77.78	28.57	88.89	35.71
afrispeech_lg_2.wav	88.89	68.85	144.44	65.57	66.67	45.90
afrispeech_rw_0.wav	72.22	50.45	55.56	26.13	55.56	43.24
afrispeech_rw_1.wav	50.00	28.00	43.75	14.67	6.25	5.33
afrispeech_rw_2.wav	85.71	53.49	57.14	43.02	78.57	50.00
afrispeech_pcm_0.wav	0.00	0.00	80.00	28.57	20.00	17.86
afrispeech_pcm_1.wav	36.36	21.11	81.82	34.44	109.09	42.22
afrispeech_pcm_2.wav	14.29	18.10	92.86	33.33	107.14	39.05

Overall mean: Sahara 41.77% WER / 22.43% CER; MMS 57.61% / 27.08%; Whisper 66.17% / 41.00%. The chart and summary values are based on these per-clip scores.

WER and CER can exceed 100% when inserted words or characters outnumber the reference units. A high rate is a scored recognition outcome, not an API failure. All baseline error fields are null.